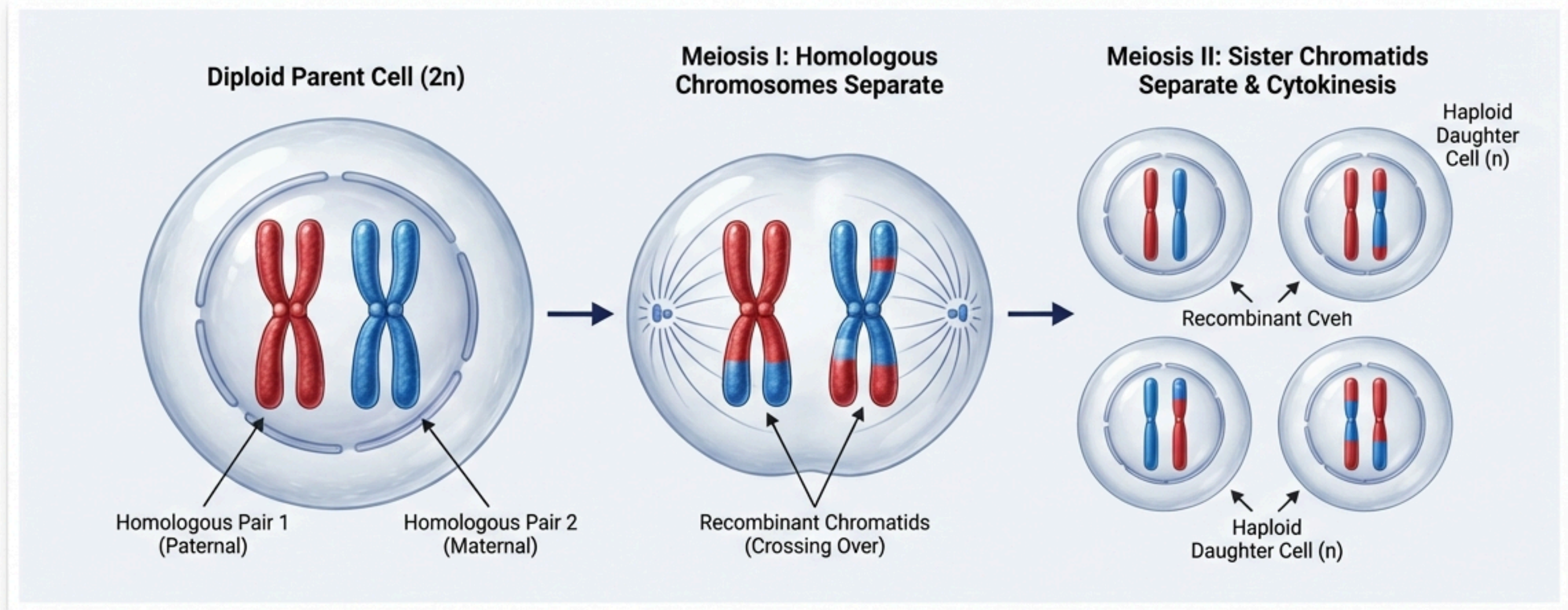


Mastering Meiosis: The Reduction Division

A High-Yield Review for NEET-UG Aspirants

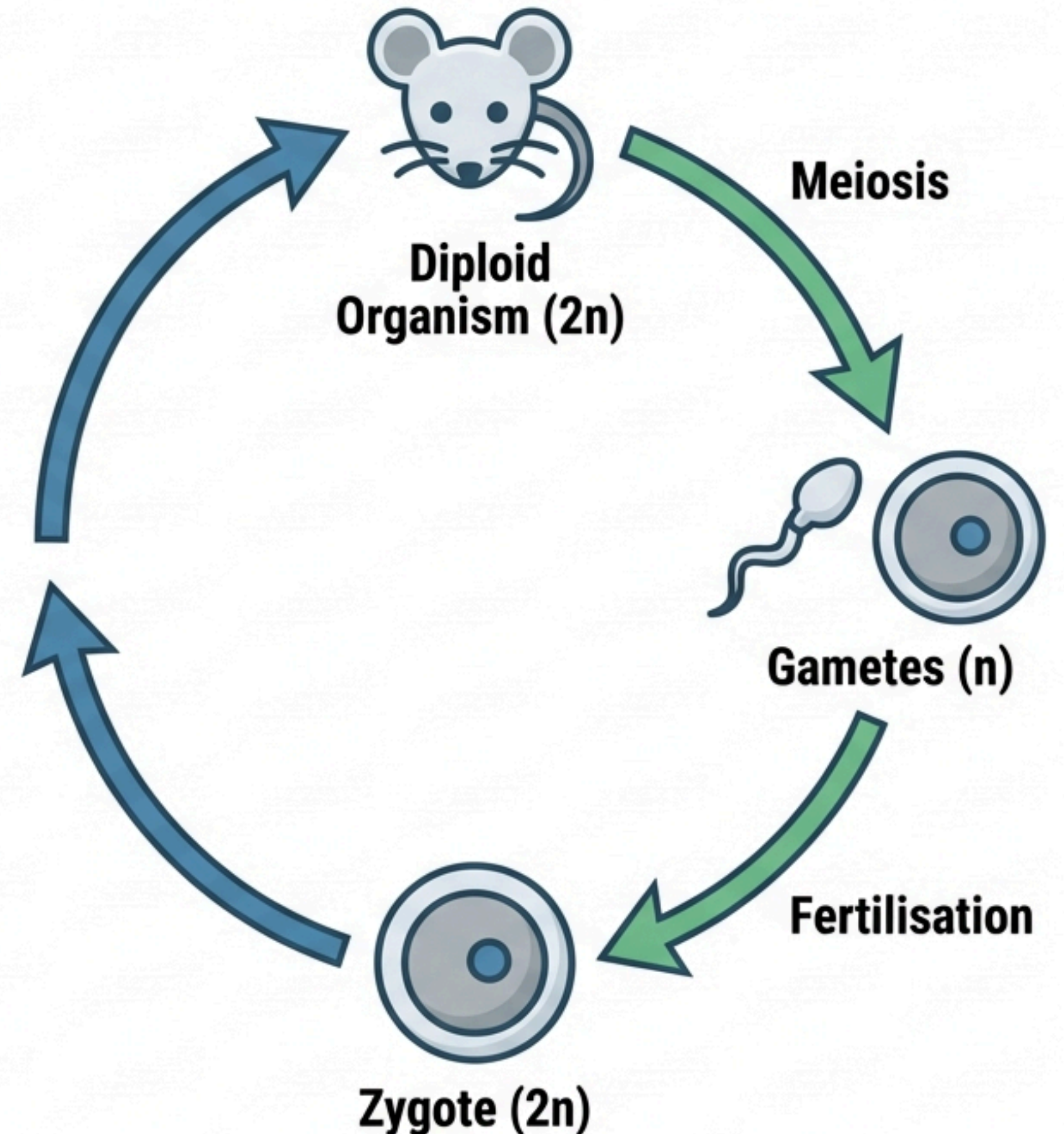


The Core Concept: Why Reduce to Survive?

- **Definition:** A specialized cell division that reduces the chromosome number by half, resulting in haploid daughter cells (n).
- **The Cycle:** Meiosis ensures the production of the **haploid phase** in the life cycle, while **fertilisation** restores the **diploid phase** ($2n$).
- **Occurrence:** Occurs during **gametogenesis** in plants and animals to form gametes.

NEET ALERT

Remember: Meiosis reduces the number ($2n \rightarrow n$) so **Fertilisation** can restore it ($n + n = 2n$).



Key Features & Rules of Division

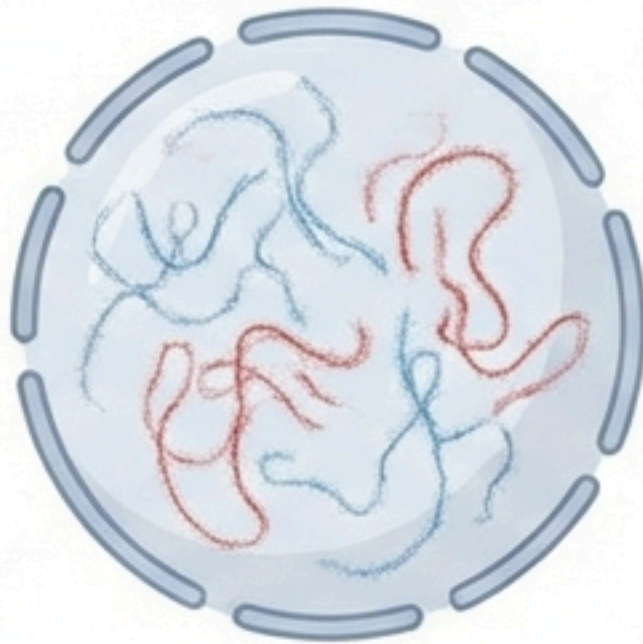


1. **Two Sequential Cycles:** Involves Meiosis I and Meiosis II.
2. **Single Replication:** Involves **only a single cycle of DNA replication.**
3. **Initiation:** Meiosis I starts *after* parental chromosomes have replicated to produce identical **sister chromatids** at the S phase.
4. **Pairing:** Involves pairing of homologous chromosomes and recombination.

Prophase I (Part 1): Structure & Synapsis

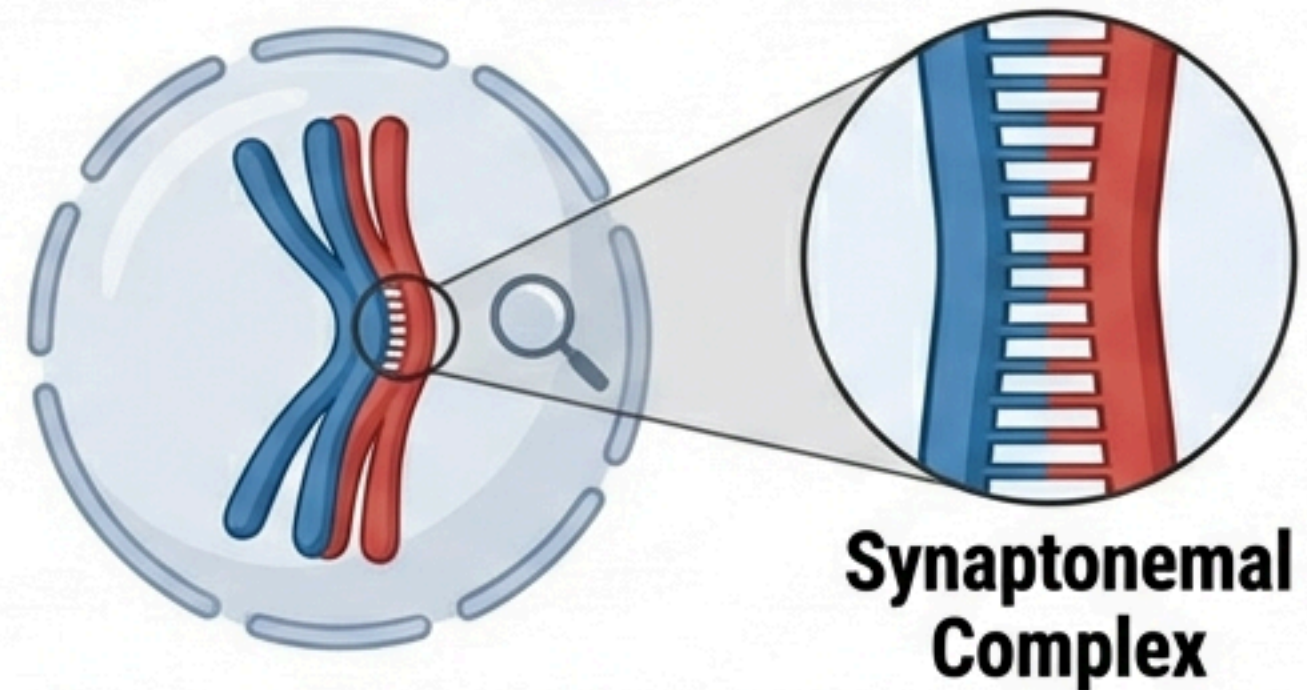
Leptotene & Zygotene

Leptotene



- Chromosomes become gradually visible under light microscope.
- **Compaction** continues throughout this stage.

Zygotene



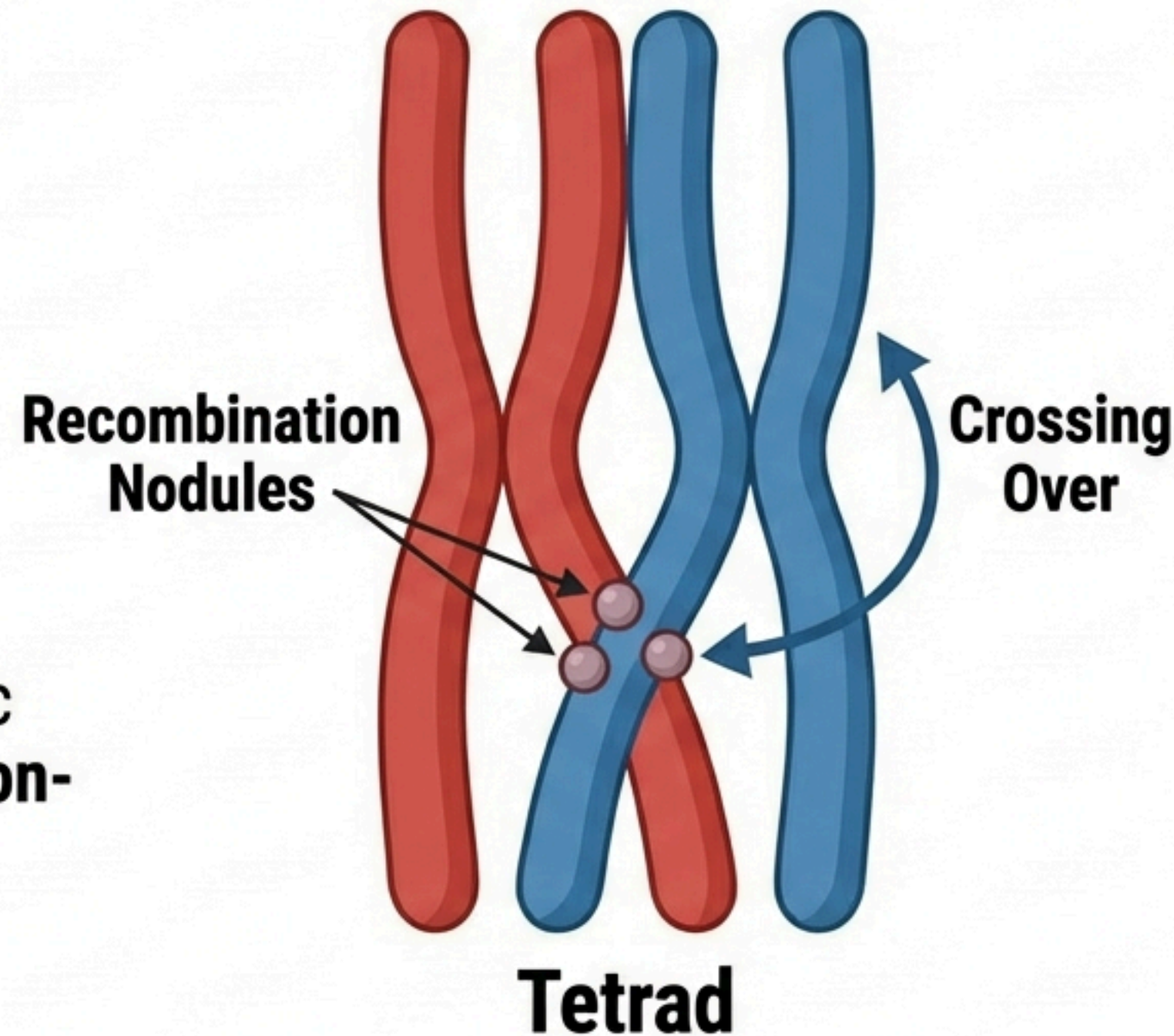
- **Synapsis:** Chromosomes start pairing together.
- Paired structures are called **Homologous Chromosomes**.
- **Synaptonemal Complex** forms between the pair.
- **Bivalent / Tetrad** formed (visible clearly in next stage).

Prophase I (Part 2): The Genetic Exchange

Pachytene

Key Characteristics:

- Bivalents clearly appear as **tetrads**.
- Appearance of **Recombination Nodules**.
- **Crossing Over**: Exchange of genetic material between **non-sister chromatids**.



Mechanism:

- Enzyme Mediated: **Recombinase**.
- Result: Recombination of genetic material is completed.



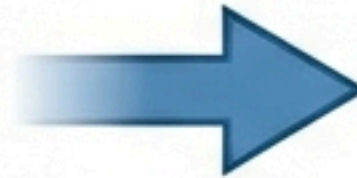
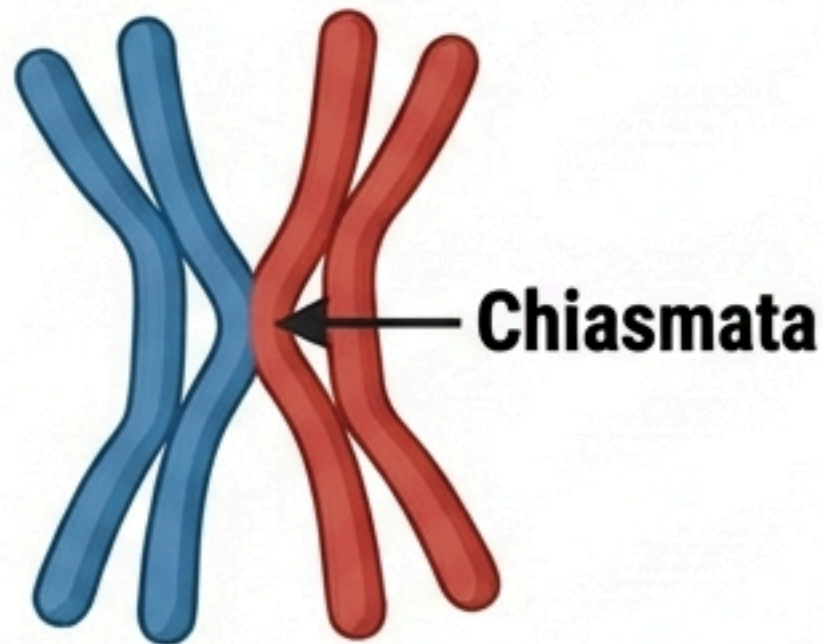
NEET ALERT

Question Trap: Crossing over happens between NON-sister chromatids, not sister chromatids.

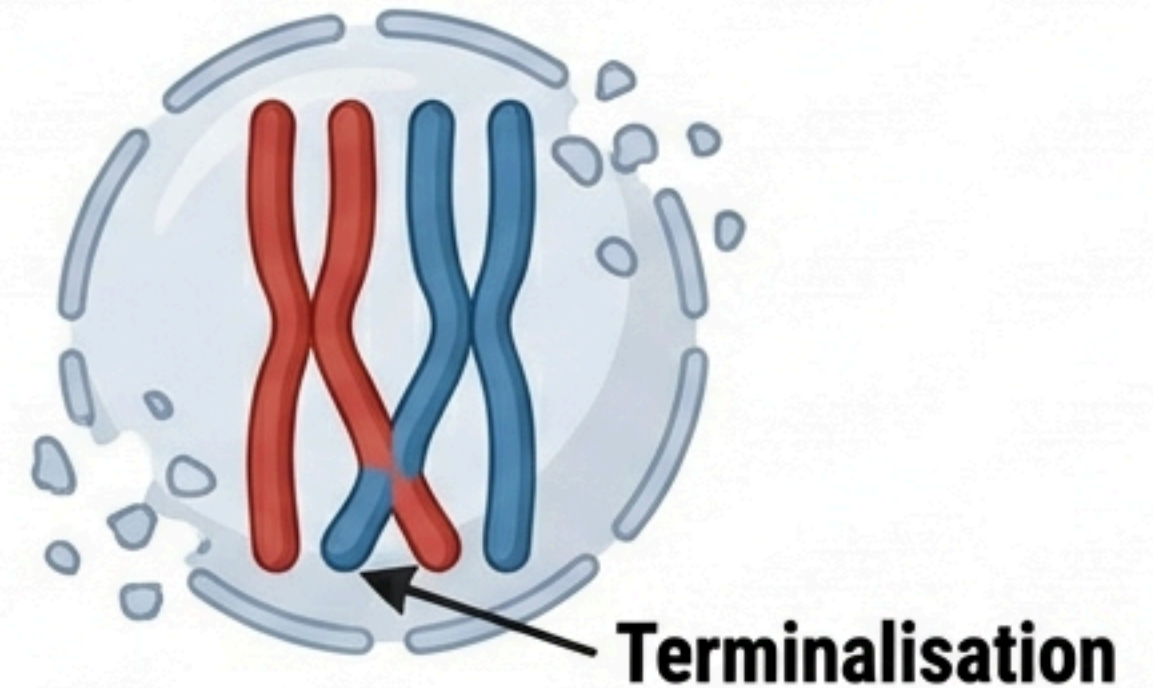
Prophase I (Part 3): Separation & Transition

Diplotene & Diakinesis

Diplotene



Diakinesis



Diplotene

- Dissolution of synaptonemal complex.
- Homologues separate but remain attached at Chiasmata.
- *Note:* Can last months/years in vertebrate oocytes.

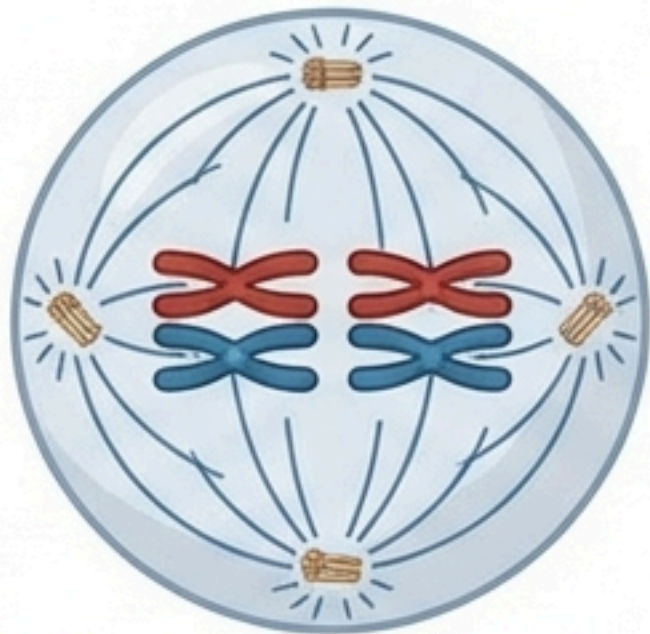
Diakinesis

- **Terminalisation** of chiasmata.
- Chromosomes fully condensed.
- Nucleolus and nuclear envelope disappear.
- Transition to Metaphase.

Meiosis I: The Reduction Step

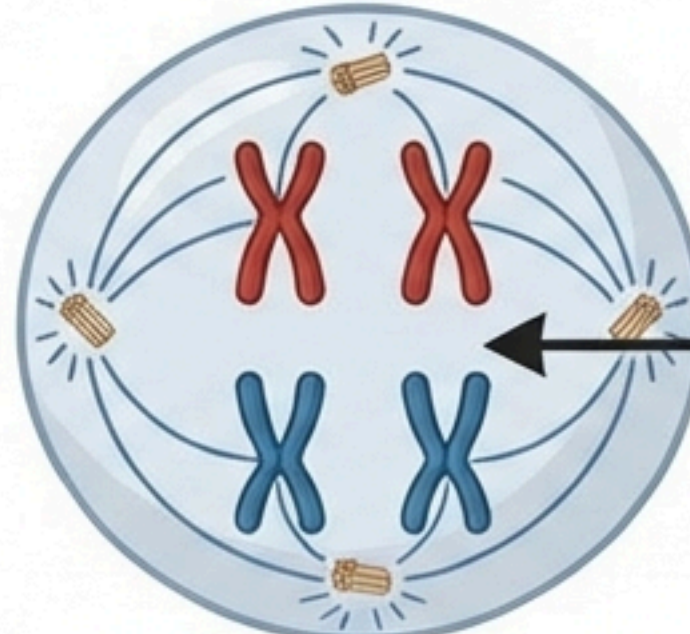
Metaphase I to Telophase I

Metaphase I



Bivalents align at equator.

Anaphase I

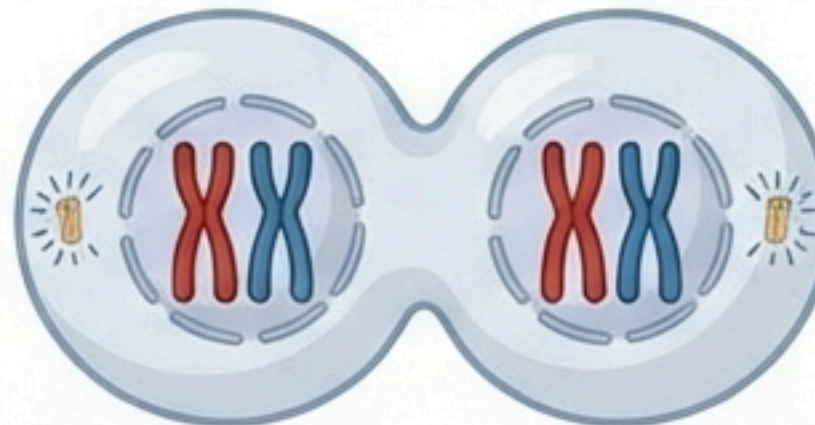


Homologous chromosomes separate.
Sister chromatids remain associated.



NEET ALERT

Anaphase I: Homologues separate. Centromeres do NOT split yet.



Telophase I

Nuclear membrane reappears.
Formation of a **Dyad of cells**.

The Interlude: Interkinesis



The stage between Meiosis I and Meiosis II.

- Generally short-lived.
- Chromosomes may disperse but do not reach extended interphase state.

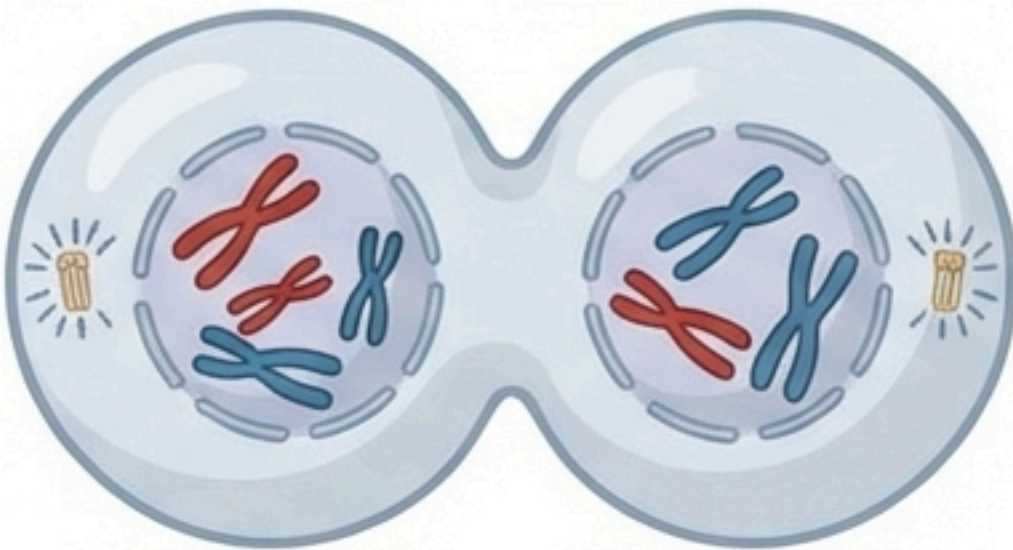
CRITICAL FACT: There is NO replication of DNA during Interkinesis.

Meiosis II: The Equational Division

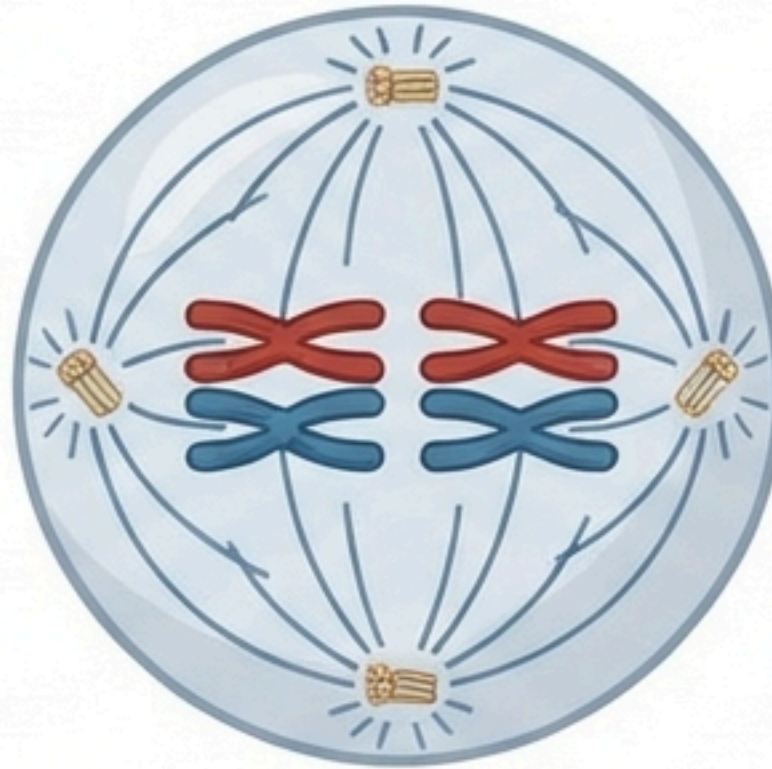
Prophase II & Metaphase II

Meiosis II resembles a **normal mitosis**. It begins immediately after cytokinesis.

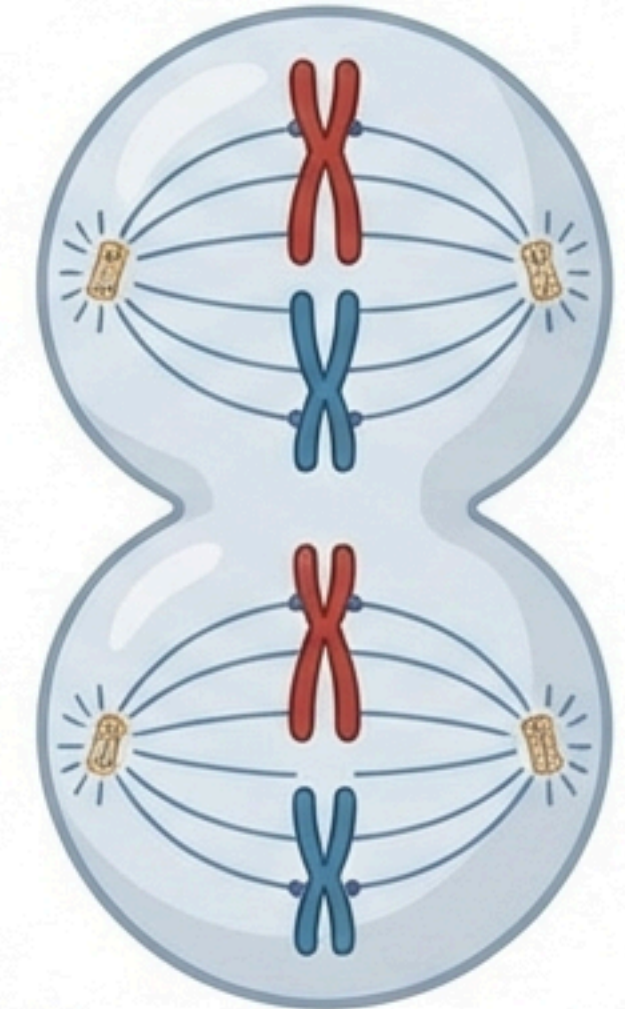
Prophase II



Recap: Metaphase I



Metaphase II

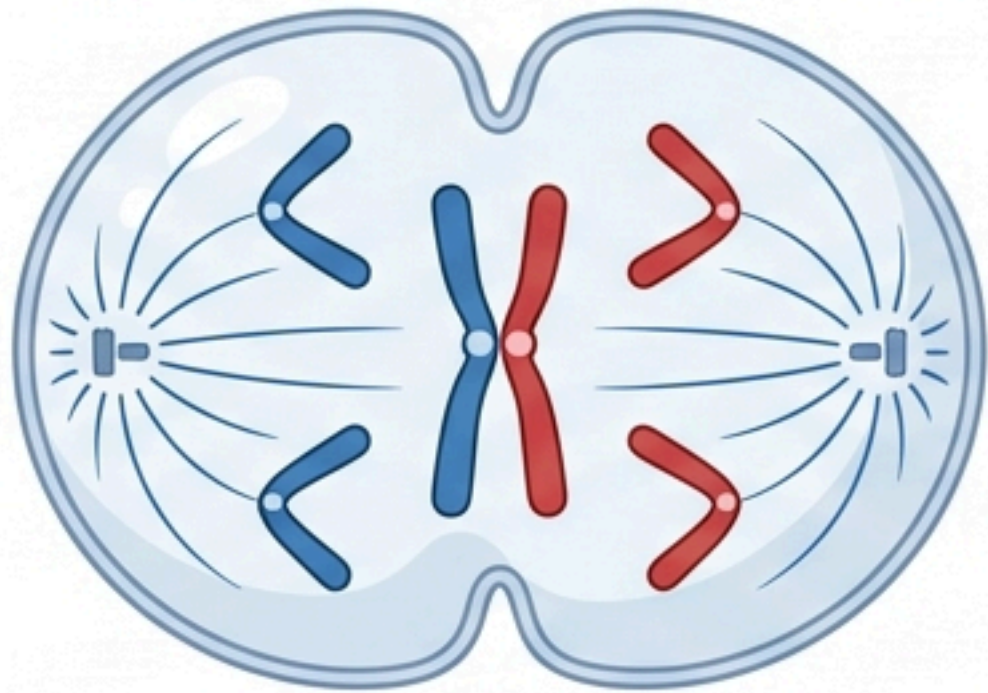


Chromosomes align at equator.
Microtubules attach to sister chromatids.

The Finale: Splitting the Centromere

Anaphase II & Telophase II

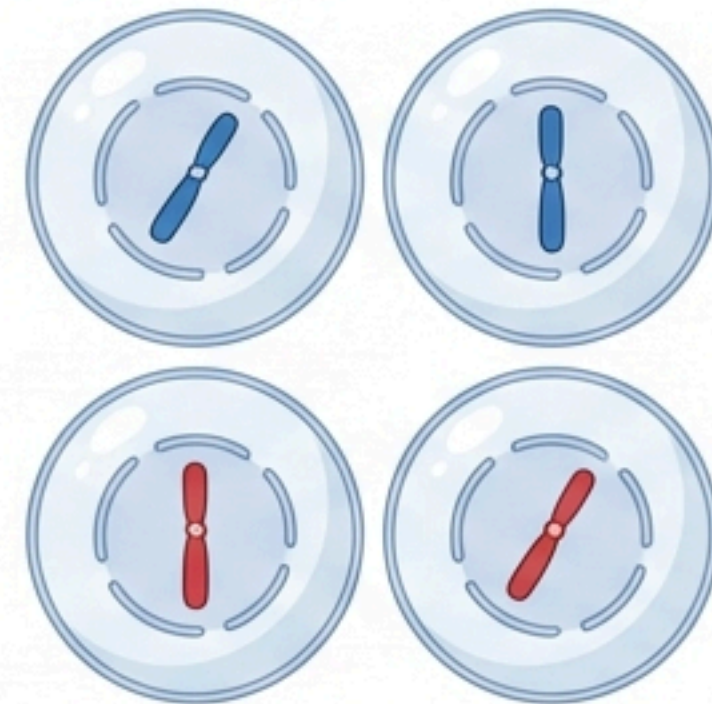
Anaphase II



Simultaneous **splitting of the centromere**. Sister chromatids move to opposite poles.



Telophase II & End Result



Nuclear envelope encloses chromosomes.

Result: Formation of a **Tetrad of cells** (4 haploid daughter cells).

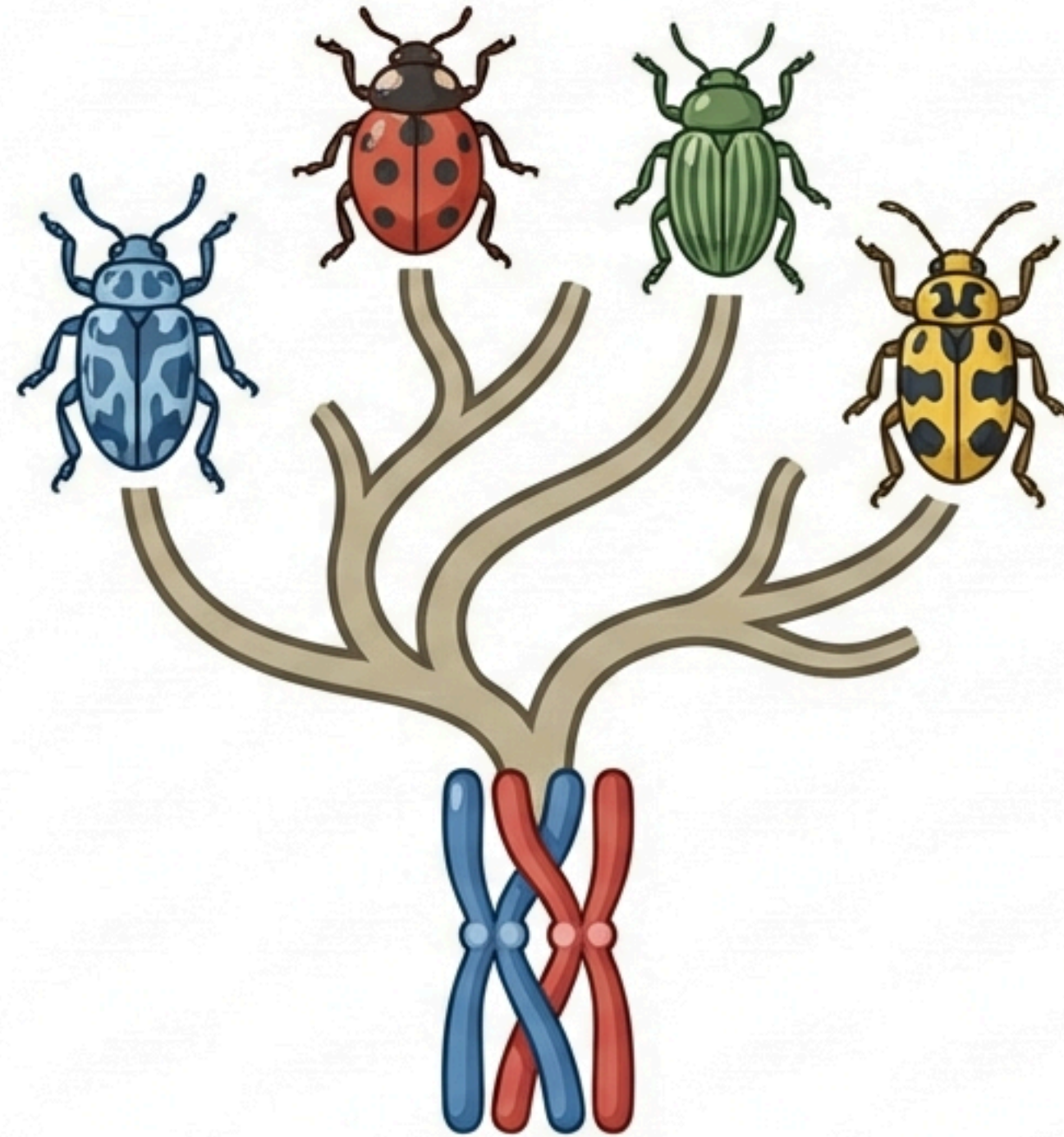
Cheat Sheet: Meiosis I vs. Meiosis II

Feature	Meiosis I	Meiosis II
Pairing	Yes (Synapsis of Homologues)	No
Metaphase Alignment	Bivalents (Pairs) at equator	Individual chromosomes at equator
Anaphase Separation	Homologous Chromosomes separate	Sister Chromatids separate (Centromere splits)
DNA Replication	Preceded by S-phase (Replication)	No replication preceding it

Significance: Evolution & Stability

Conservation:

Paradoxically, Meiosis reduces chromosome number by half to ensure the specific chromosome number is conserved across generations ($n + n = 2n$).



Variation: Crossing over **increases genetic variability** in the population from one generation to the next.

Evolution: These variations are very important for the process of evolution.

NCERT 'Golden Lines' to Memorize

Direct quotes frequently tested in True/False questions.

Meiosis ensures the production of haploid phase... whereas **fertilisation restores the diploid phase.**

Meiosis involves two sequential cycles... but **only a single cycle of DNA replication.**

Crossing over is an enzyme-mediated process and the enzyme involved is called **recombinase.**

In oocytes of some vertebrates, **diplotene can last for months or years.**

Interkinesis... there is **no replication of DNA.**

NEET Practice Challenge (Part 1)

Q1: During which stage of Prophase I does the synaptonemal complex dissolve?	(A) Pachytene (B) Zygotene (C) Diplotene (D) Leptotene
Q2: The enzyme Recombinase is involved in:	(A) Metaphase I (B) Anaphase II (C) Pachytene (D) Diakinesis
Q3: Synapsis occurs between:	(A) Two homologous chromosomes (B) A male and female gamete (C) mRNA and Ribosomes (D) Spindle fibers and centromeres
Q4: Terminalisation of chiasmata occurs in:	(A) Diplotene (B) Diakinesis (C) Pachytene (D) Metaphase I
Q5: Which stage is characterized by the appearance of recombination nodules?	(A) Leptotene (B) Zygotene (C) Pachytene (D) Diplotene

NEET Practice Challenge (Part 2)

Q6: The splitting of the centromere occurs during:	(A) Anaphase I (B) Anaphase II (C) Metaphase II (D) Telophase I
Q7: Identify the correct statement regarding Interkinesis:	(A) It is long-lived (B) DNA replication occurs (C) It is followed by Prophase II (D) It creates a tetrad of cells
Q8: Bivalents align on the equatorial plate during:	(A) Metaphase I (B) Metaphase II (C) Anaphase I (D) Diakinesis
Q9: Meiosis I ends with the formation of:	(A) Four haploid cells (B) Two diploid cells (C) A dyad of cells (D) A tetrad of cells
Q10: In Meiosis, how many cycles of DNA replication occur?	(A) One (B) Two (C) Four (D) Zero

Answer Key: 1:C, 2:C, 3:A, 4:B, 4:B, 5:B, 7:A 5:C, 8:A, 9:C, 10:A